

1. Definitions and characteristics of system & state functions

Open system (like a coffee cup): matter and energy can enter or leave

Closed system (like an unopened bottle): only energy can enter or leave

Isolated system (like a thermos): matter and energy cannot enter or leave

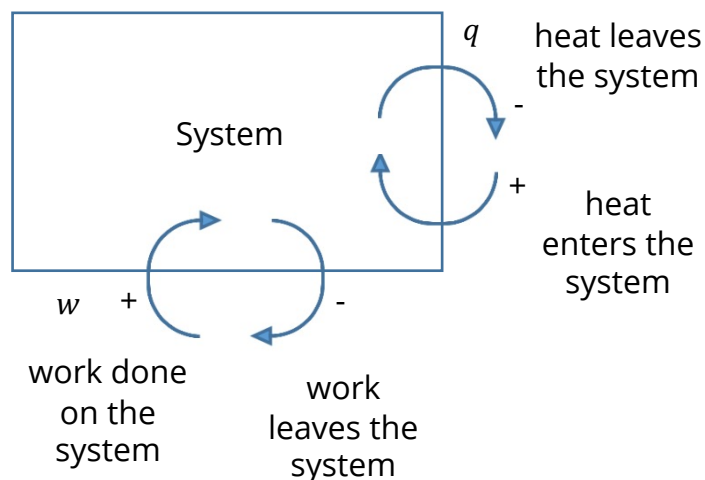
2. Sign conventions: ΔG , ΔH , ΔS , w , q

$$w = -\Delta nRT = -P\Delta V$$

as volume increases w is (-)

as volume decreases w is

(+)



1. Definitions and characteristics of system & state functions

State Functions:

$\Delta U = \Delta E$ the change in internal energy

ΔH the measure of heat change

ΔS the measure of disorder

ΔG the free energy of the system

Path Functions:

q and w are path functions, not state functions. They describe where energy is being expended

2. Sign conventions: ΔG , ΔH , ΔS , w , q

Understand sign conventions

ΔG (-) spontaneous
(+) nonspontaneous

ΔH (-) exothermic
(+) endothermic

ΔS (+) increased disorder
(-) increased order

w (+) work done on system
(-) work done on surroundings

q (+) heat added to system
(-) heat added to surroundings